Mathematical Model & `Reactome` quirks

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Contents

1 Kinetic constants estimation

Notation preliminaries:

- $\mathbb{N}_1 = \mathbb{N} - \{0\}$
- $\mathbb{R}^+ = \{x \mid x \in \mathbb{R} \land x \geq 0\}$

1.1 Model definition 4

Let $\mathcal{T} = \{\texttt{reactant}, \texttt{product}, \texttt{enzyme}, \texttt{activator}, \texttt{inhibitor}\}$ be the set of roles an entity can undertake in a reaction. In order to simplify notation, the following subsets of $\mathcal{T}$ shall be defined:

- $\mathcal{T}_{\texttt{stoichiometric}} = \{\texttt{reactant}, \texttt{product}\}$ the set of stoichiometric roles
- $\mathcal{T}_{\texttt{modifier}} = \{\texttt{enzyme}, \texttt{activator}, \texttt{inhibitor}\}$ the set of modifier roles
- $\mathcal{T}_{\texttt{reagent}} = \{\texttt{reactant}\} \cup \mathcal{T}_{\texttt{modifier}}$ the set of reagent (input) roles

Definition 1 *(Biological network).*
A biological network $\mathrm{G} = (\mathrm{V}, \mathrm{E}, \mathcal{c}, n)$ is a semi-quantitative graph model where

- $\mathrm{V} = \mathrm{X} \sqcup \mathrm{R} \sqcup \mathrm{C}$
  - X is the set of physical entities
  - R is the set of reactions
  - C is the set of compartments in which the entities are located
- $\mathrm{E} = \mathrm{E}_{\texttt{stoichiometric}} \sqcup \mathrm{E}_{\texttt{modifier}}$
  - $\mathrm{E}_{\texttt{stoichiometric}} \subseteq \mathrm{X} \times \mathrm{R} \times \mathcal{T}_{\texttt{stoichiometric}}$
  - $\mathrm{E}_{\texttt{modifier}} \subseteq \mathrm{X} \times \mathrm{R} \times \mathcal{T}_{\texttt{modifier}}$
- $\mathcal{c} : \mathrm{X} \to \mathrm{C}$ is a function that assigns to each entity a compartment
- $n : \mathrm{E}_{\texttt{stoichiometric}} \to \mathbb{N}_1$ is the stoichiometry function

Definition 2 *(Scope).*
Given a biological network G let $\mathrm{S} = (\mathrm{X}_{\texttt{interest}}, \mathrm{X}_{\texttt{boundary}}, \mathrm{R}_{\texttt{interest}})$ be the scope of a problem in G, where

- $\mathrm{X}_{\texttt{interest}} \subseteq \mathrm{X}$ is the set of entities whose reachability must be computed
- $\mathrm{X}_{\texttt{boundary}} \subseteq \mathrm{X}$ is the set of entities on the boundary
- $\mathrm{R}_{\texttt{interest}} \subseteq \mathrm{R}$ is the set of the only reaction that can be included in the reachability of $\mathrm{X}_{\texttt{interest}}$

Definition 3 *(Scope reachability).*
Given a biological network G and a scope S in G let $\mathrm{G}' = (\mathrm{V}', \mathrm{E}', \mathcal{c}', n')$ be the subgraph of G reachable from S.

The set of reactions R' in G' is defined inductively as follows:

- if x is an entity and r is a reaction where $x \in \mathrm{X}_{\texttt{interest}} \land r \in \mathrm{R}_{\texttt{interest}} \land (x, r, \texttt{product}) \in \mathrm{E}$ then $r \in \mathrm{R}'$
- if r_1 is a reaction in R', r_2 is a reaction in $\mathrm{R}_{\texttt{interest}}$ and there is some entity x where $x \notin \mathrm{X}_{\texttt{boundary}}$ and $\{(x, r_1)\} \times \mathcal{T}_{\texttt{reagent}} \cap \mathrm{E} \neq \varnothing$ and $(x, r_2, \texttt{product}) \in \mathrm{E}$ then $r_2 \in \mathrm{R}'$

- $\mathrm{X}' = \{x \mid \exists\, r, t \;\; r \in \mathrm{R}' \wedge (x, r, t) \in \mathrm{E}_{\mathrm{R}}\}$
- C' is the image of $\mathscr{c}$ over X'
- $\mathrm{E}' = \{(x, r, t) \mid r \in \mathrm{R}' \wedge (x, r, t) \in \mathrm{E}\}$
- $\mathscr{c}'$ is just like $\mathscr{c}$ but restricted to X'
- n' is just like n but restriceted to $\mathrm{E}'_{\texttt{stoichiometric}}$

1.2 Reactions rate laws 2

$$\texttt{regulation}(r) = \prod_{(x,r,\ \texttt{activator})\ \in\ \mathrm{E}} \frac{[x]}{K_A^{x,r} + [x]} \prod_{(x,r,\ \texttt{inhibitor})\ \in\ \mathrm{E}} \frac{K_I^{x,r}}{K_I^{x,r} + [x]} \tag{1}$$

$$\texttt{rate}(r) = \texttt{regulation}(r) \cdot k_r \prod_{\mathrm{e}=(x,r,\ \texttt{reactant})\ \in\ \mathrm{E}} [x]^{n(\mathrm{e})} \tag{2}$$

$$\texttt{rate}(r) = [e] \cdot \texttt{regulation}(r) \cdot \frac{N}{D} \tag{3}$$

$$N = k_r \prod_{\mathrm{e}=(x,r,\ \texttt{reactant})\ \in\ \mathrm{E}} \left(\frac{[x]}{K_m^{x,r}}\right)^{n(\mathrm{e})} \tag{4}$$

$$D = \text{to define...} \tag{5}$$

In the “Systems Biology” book (Klipp, Edda, Kowald, Axel, pages 49-50) it says that the the denominator D can be one of the following:

1. Power-law modular rate law: $D = 1$
2. Common modular rate law (Equation 5)
3. Simultaneous binding modular rate law

$$D = \prod_{\substack{(x,\ \texttt{reactant},n) \\ \in \\ f_{\mathrm{R}}(r)}} \left(1 + \frac{[x]}{K_m^{x,r}}\right)^n \prod_{\substack{(x,\ \texttt{product},n) \\ \in \\ f_{\mathrm{R}}(r)}} \left(1 + \frac{[x]}{K_m^{x,r}}\right)^n \tag{6}$$

4. Direct binding modular rate law:

$$D = 1 + \prod_{\substack{(x,\ \texttt{reactant},n) \\ \in \\ f_{\mathrm{R}}(r)}} \left(\frac{[x]}{K_m^{x,r}}\right)^n + \prod_{\substack{(x,\ \texttt{product},n) \\ \in \\ f_{\mathrm{R}}(r)}} \left(\frac{[x]}{K_m^{x,r}}\right)^n \tag{7}$$

5. Force-dependent modular rate law:

$$D = \sqrt{\prod_{\substack{(x,\ \texttt{reactant},n) \\ \in \\ f_{\mathrm{R}}(r)}} \left(1 + \frac{[x]}{K_m^{x,r}}\right)^n \prod_{\substack{(x,\ \texttt{product},n) \\ \in \\ f_{\mathrm{R}}(r)}} \left(1 + \frac{[x]}{K_m^{x,r}}\right)^n} \tag{8}$$

Definition 4 *(Biological model satisfiability problem).*
Given a biological network G let T, I be such that:

- T is the time horizon for the simulation
- TODO: tollerances for kinetic constants of the same reaction
- TODO: tollerances for kinetic constants of reversible reactions
- I is the set of indicators to measure
  - given a trajectory
- TODO: additional constrain

Let $\langle \theta, D, C \rangle$ be the satisfiability problem where

$$\begin{aligned}\theta = & \{t_r \mid r \in \mathrm{R}\} \\ & \cup \{t_x \mid x \in \mathrm{X}\} \\ & \cup \{t_A^{x,r} \mid (x, r, \texttt{activator}) \in \mathrm{E}\} \\ & \cup \{t_I^{x,r} \mid (x, r, \texttt{inhibitor}) \in \mathrm{E}\} \\ & \cup \left\{t_{m,+}^{x,r} \mid r \text{ involves an enzyme and } x \text{ is a reactant of } r\right\} \\ & \cup \left\{t_{m,-}^{x,r} \mid r \text{ involves an enzyme and } x \text{ is a product of } r\right\} \\ & \cup \left\{x_{\text{avg}} \mid x \in \mathrm{X}\right\}\end{aligned} \tag{9}$$

The respsective kinetic constants are defined as 10^t, for example

- $k_A^{x,r} = 10^{t_A^{x,r}}$

$$\begin{aligned}D = & \left\{D_{t_x} \mid D_{t_x} = [0, 1]\right\} \\ & \cup \{\text{all other domains are } [-6, 6]\}\end{aligned} \tag{10}$$

Constraints

$$\forall x \in \mathrm{X} \quad 0 \leq [x] \leq 1 \tag{11}$$

$$\forall x \in \mathrm{X}_{\text{avg}} \quad [x](\varphi \cdot T) - [x](T) = 0 \text{ (stability)} \tag{12}$$

$$\forall x_{\text{avg}} \in \mathrm{X}_{\text{avg}} \quad \left[x_{\text{avg}}\right] \in [\mathrm{v}(x) - \varepsilon, \mathrm{v}(x) + \varepsilon] \tag{13}$$

$$\begin{aligned}& \forall k_1^r, k_2^r \in X, x \in \mathrm{X} \\ & \quad \text{if } \ x \ \text{ is a modifier of } \ k_{r_2} \text{ and is produced by } k_1^r \text{ then } k_{r_1} < k_{r_2}\end{aligned} \tag{14}$$

Definition 5 *(Optimization problem).*
Basically take the satisfiability problem above and turn it in a "single objective optimization problem"